These equations use hyperbolic sine and cosine functions to define the ellipse, just as ordinary sine and cosine functions operate on a circle. The flatness of the ellipse is controlled by the variable: *PR*, which is numerically equal to the percentage of ripple in the filter's passband. The variables: ϵ, v and k are used to reduce the complexity of the equations, and are represented in the program by: *ES*, *VX* and *KX*, respectively. In addition to converting from a circle to an ellipse, these equations correct the pole locations to keep a unity cutoff frequency. Since many programming languages do not support hyperbolic functions, the following identities are used:

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\sinh^{-1}(x) = \log_e [x + (x^2 + 1)^{1/2}]$$

$$\cosh^{-1}(x) = \log_e [x + (x^2 - 1)^{1/2}]$$

These equations produce illegal operations for $PR \geq 30$ and $PR = 0$. To use this program to calculate Butterworth filters (i.e., zero ripple, $PR = 0$), the program lines that implement these equations must be bypassed (line 1120).

Continuous to Discrete Conversion

The most common method of converting a pole-zero pattern from the s-domain into the z-domain is the **bilinear transform**. This is a mathematical technique of *conformal mapping*, where one complex plane is algebraically distorted or warped into another complex plane. The bilinear transform changes $H(s)$, into $H(z)$, by the substitution:

EQUATION 33-10
The Bilinear transform. This substitution maps every point in the s-plane into a corresponding piont in the z-plane.

$$s \rightarrow \frac{2(1 - z^{-1})}{T(1 + z^{-1})}$$

That is, we write an equation for $H(s)$, and then replaced each s with the above expression. In most cases, $T = 2\tan(1/2) = 1.093$ is used. This results in the s-domain's frequency range of 0 to π radians/second, being mapped to the z-domain's frequency range of 0 to π radians. Without going into more detail, the bilinear transform has the desired properties to convert